

One-Page Plan: Log Data Analytics (ML-Focused)

1. Core Concept & Data Nature

Log data is inherently **sequential** and contains **timestamps**, which makes it ideal for time-series analysis.

- **Key Log Elements:** Errors, Anomalies, Root cause of malfunctions, and System resource consumption.
- **Statistical Foundation:** Descriptive analytics (Mean, Median, Mode, SD, Variance) are used to establish a "baseline" for normal system behaviour.

2. Pre-processing & Data Integrity (The "ML-Seq" Pipeline)

Before feeding logs into a model, the data must be cleaned and "smoothed."

- **Handling Nulls/NaNs:** Distinguishing between Zero vs. Null vs. NaN is critical for **Conditional Probability**. Use **Imputation Techniques** to fill gaps.
- **Smoothing:** Necessary to remove "noise" from audio/video logs or system spikes.
- **Vocabulary:** Handling **OOV (Out of Vocabulary)** tokens in text-based logs and addressing **Recurrence Frequency** (e.g., repeating transaction logs).

3. Advanced Analytics & Modelling

The notes highlight a transition from simple statistics to complex neural architectures:

- **Anomaly & Outlier Detection:**
 - **Statistical:** Using Z-Score and **IQR (Inter-Quartile Range)** to set thresholds.
 - **Behavioural:** Detecting "Doom scrolling," Unauthorized Access, or sudden spikes in CPU utilization.
- **Algorithms:**
 - **Clustering:** K-Means, Hierarchical, and **DBSCAN** for identifying "Unknown Anomalies."
 - **Sequential Models:** Since timestamps have a sequential nature, use **LSTM (Long Short-Term Memory)** or RNNs.
 - **Probabilistic:** **HMM (Hidden Markov Models)**—widely used in finance and sequence prediction.
- **Neural Networks:** Used for "Crazy non-linearity" where simple statistical methods fail $X \rightarrow y'$.

4. Domain-Specific Applications

- **Security (IDS):** Intrusion Detection Systems to stop DoS (Denial of Service) attacks.
- **Video/Audio Analytics:** Addressing "Bad Precision" by resizing to lower resolutions or retraining on high-res data to improve user experience.

- **Finance:** Using HMM for pattern recognition in transactional log sequences.

Implementation Steps:

1. **Data Collection Phase:** Ingest logs, focusing on timestamps and duplicates.
2. **Cleaning:** Create a data frame, remove duplicates, and perform data imputation.
3. **Feature Engineering:** Perform clustering and dimensionality reduction on high-volume data.
4. **Normalization:** The process of scaling numerical log features (like latency or CPU use) to a standard range (e.g., 0 to 1) to ensure the model isn't biased by different units of measurement.
5. **Model Training (LSTM/RNN):** Teaching a deep learning architecture to learn the "normal" sequential patterns and dependencies of log events over time using historical timestamp data.
6. **Threshold Setting (Z-Score):** Defining a mathematical boundary typically based on standard deviations from the mean to distinguish between "usual" system behaviour and a statistical outlier.
7. **Anomaly Alerting:** The final operational step where the system triggers a notification or "flag" when the model's prediction error or a specific metric crosses the predefined threshold.